

INTRODUCTION

This chapter will introduce the basic concepts and definitions relating to computer programming. The reader already familiar with these concepts may want to glance quickly at the contents of this chapter and then move on to Chapter 2. It is suggested, however, that even the experienced reader look at the contents of this introductory chapter. Many significant concepts are presented here including, for example, two's complement, BCD, and other representations. Some of these concepts may be new to the reader; others may improve the knowledge and skills of experienced programmers.

WHAT IS PROGRAMMING?

Given a problem, one must first devise a solution. This solution, expressed as a step-by-step procedure, is called an *algorithm*. An algorithm is a step-by-step specification of the solution to a given problem. It must terminate in a finite number of steps. This algorithm may be expressed in any language or symbolism. A simple example of an algorithm is:

- 1—insert key in the keyhole
- 2—turn key one full turn to the left
- 3—seize doorknob
- 4—turn doorknob left and push the door